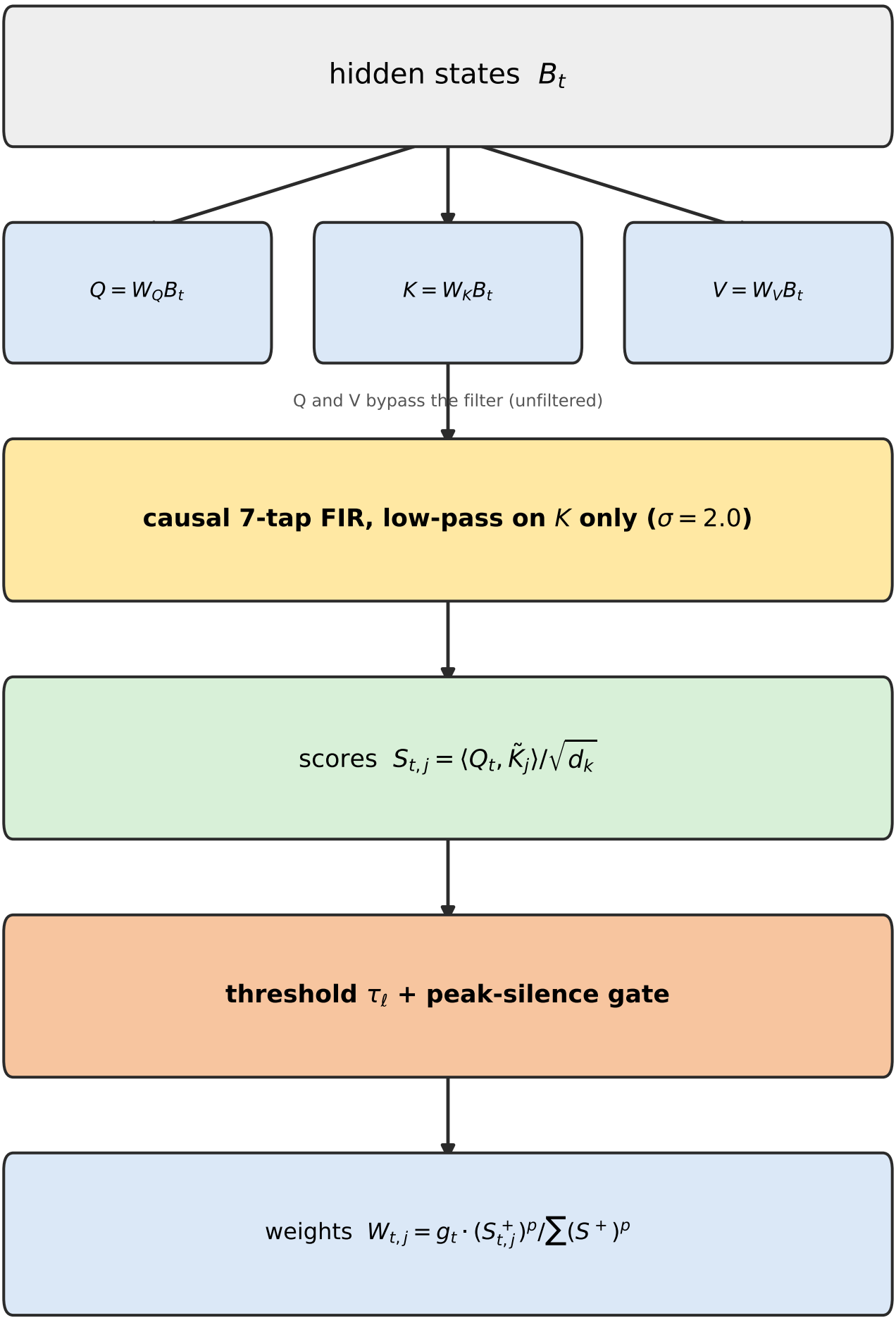
pn_fir_tau:
causal FIR on K + a calibrated off switch

The filter acts on the key stream only; Q and V stay sharp.



What this buys you

One layer binds a queried key to the adjacent value slot.
Non-bijective recall: 44.5% → 100.0%

Fixed $\tau = 0$ rectification starves: 64.4% recall,
91.3% negative scores, 44.5% dead rows.

One learned τ per layer: 100% recall; unanswerable row mass
0.002 vs 0.100 for virtual sinks.